Silicon ball lightning: the chemical model and the welding arc

Can burning silicon nanoparticles make a ball that lasts seconds, and can a welding arc make them?

kronspice simulation report · 2026-11-08 · about 4 seconds; deterministic

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Summary

Abrahamson and Dinniss (Nature 403, 519, 2000) propose that lightning striking soil reduces its silica and ejects a network of silicon nanoparticles whose slow oxidation in air gives a ball of light lasting seconds. Cen, Yuan and Xue (Phys. Rev. Lett. 112, 035001, 2014) recorded the first spectrum of a natural ball: silicon, iron and calcium lines from the soil throughout its 1.64 s. This example tests the chemical model's energy and timing, simulates a welding arc at the operating point of the laboratory silicon balls, and simulates the 35 kV power line beside the Qinghai ball, all with kronspice and its energy ledger.

- **Energy:** burning silicon releases 32.4 kJ/g, 1.84 times the 17.6 kJ/g it costs to vaporize; an average ball (100 W for 10 s) needs only 31 mg. Iron returns 0.89 of its cost, manganese 1.24, calcium 3.08. Oxides (sand, rutile flux) store nothing.
- **Timing:** Deal–Grove kinetics burn 5–100 nm silicon particles in minutes to days at 1100–1500 K; 10 nm particles burn in 1.6–10 s only at about 2200–2900 K (extrapolated), consistent with the 3140 K laboratory estimate.
- **Welding arc (kronspice):** 70 V open circuit behind 1.447 mH gives 120 A rms at 23 V, 2.5 kW, ledger 2×10^{-5}; 1–10 % of one second of arc vaporizing silicon stores 46–460 J, enough for a 100 W ball of 0.5–4.6 s.
- **Power line (kronspice):** the line's coupling makes any power in the ball pulse at 100 Hz (observed light 99.4 Hz) but delivers at most 60 μW to a 15 cm ball against the 100 W it shines: it can flicker the ball, not power it.

Results

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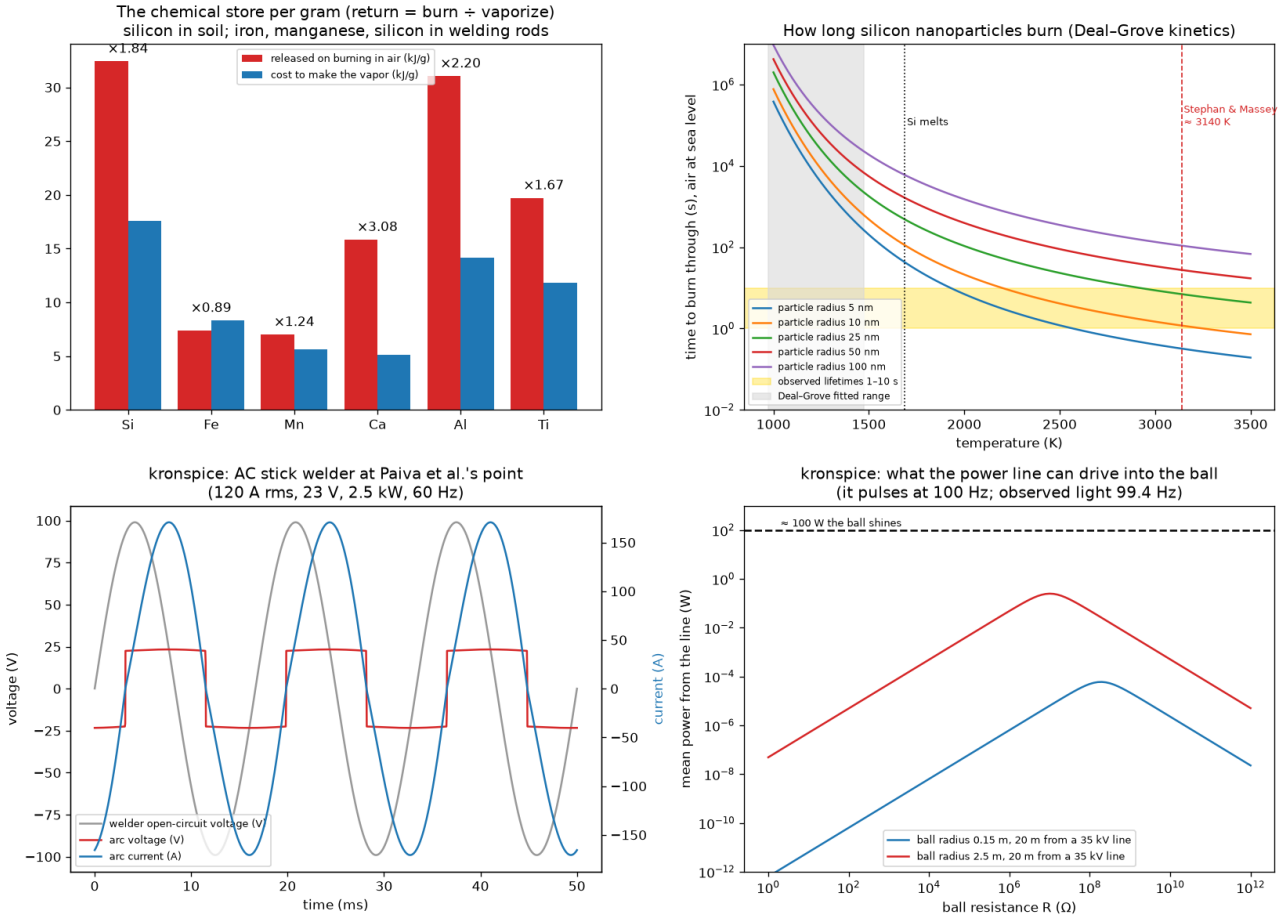


Figure 1. Top left: heat released on burning against the cost to vaporize, per gram. Top right: time for silicon particles to burn through (Deal–Grove, air), against the observed 1–10 s; grey, the measured range of the kinetics. Bottom left: the kronspice welder and arc. Bottom right: power the 35 kV line can drive into a ball 20 m away, against the 100 W it shines.

1. The chemical store

Element → oxide	Burns (kJ/g)	Vaporize (kJ/g)	Return	Mass for 1 kJ
Si → SiO_2	32.43	17.60	1.84	31 mg
Fe → Fe_2O_3	7.38	8.32	0.89	136 mg
Mn → MnO	7.01	5.65	1.24	143 mg
Ca → CaO	15.84	5.14	3.08	63 mg
Al → Al_2O_3	31.05	14.14	2.20	32 mg
Ti → TiO_2	19.72	11.79	1.67	51 mg

Soil holds silicon as SiO_2: the strike must first reduce it, paying the same 32.4 kJ/g, so the ball releases energy the lightning stored. In a welding rod the steel core returns less than it costs; the silicon and manganese deoxidizers return more.

2. How fast the particles burn

Deal and Grove's dry-oxidation constants for silicon (B = 772 μm²/h·exp(−1.23 eV/kT), B/A = 6.23×10^6 μm/h·exp(−2.00 eV/kT), J. Appl. Phys. 36, 3770, 1965), scaled by the oxygen partial pressure of air, with a particle burned through when its shell has consumed its radius, give the temperature at which a particle burns through in the observed lifetime:

Particle radius	1.64 s	3.5 s	10 s
10 nm	2940 K	2568 K	2198 K
25 nm	4582 K	3694 K	2917 K
50 nm	above 5000 K	above 5000 K	4030 K

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At the 1100–1500 K where the kinetics were measured the same particles take minutes to days. Seconds need a hot ball (2000–3000 K) of fine particles, which the laboratory estimate (3140 K) and the white colour of the Qinghai ball support; above 1473 K these values are extrapolations.

3. The welding arc in kronspice

An AC stick welder is a transformer whose leakage inductance sets the current; the arc is a current-controlled element whose voltage stays near 22.5 V. With 70 V open circuit, 1.447 mH gives 120.0 A rms at 23.0 V rms and 2494 W, inside Paiva et al.'s 20–25 V, 100–140 A, 60 Hz; the ledger closes to 2.1×10^{-5}.

Metal	Arc fraction vaporizing	Vapor	Chemical store	100 W ball lasts
Si	1 % / 10 %	1.4 / 14 mg	46 / 460 J	0.5 / 4.6 s
Fe	1 % / 10 %	3.0 / 30 mg	22 / 221 J	0.2 / 2.2 s
Mn	1 % / 10 %	4.4 / 44 mg	31 / 310 J	0.3 / 3.1 s

4. The power line beside the Qinghai ball

A 35 kV line (20.2 kV to ground, 50 Hz) 20 m away couples through a mutual capacitance $4\pi\varepsilon_0 a^2/d$; the ball has $4\pi\varepsilon_0 a$ to ground. The power into its resistance pulses at 100.00 Hz in kronspice (light observed at 99.4 Hz). The most it can be is the matched-load value $\tfrac{1}{4}\omega C_m{}^2V^2/(C_m + C_b)$: 6.0×10^{-5} W for a 15 cm ball, 0.25 W even for the full 5 m luminous size, against 100 W of light.

Ledger and validation

- Welder: kronspice ledger mismatch 2.1×10^{-5} of the source energy.
- Power line: the sweep maximum equals the matched-load formula to 2 % (TestLine).
- Deal–Grove B at 1473 K = 0.0477 μm²/h, the textbook value (TestDealGrove).
- Thermochemistry: NIST-JANAF / CRC values; vaporization cost uncertain by about ±10 % (mean heat capacities).

Limits of the model

- Deal–Grove is planar-wafer kinetics measured at 973–1473 K; nanoparticles and higher temperatures are extrapolations.
- The fraction of arc energy that vaporizes the electrode is declared (1–10 %), not computed.
- The ball's radiation, convection, buoyancy and particle network structure are not modelled.
- The line-coupling capacitances are point-sphere estimates.

Status

The chemical model passes the energy test easily and the timing test if the ball is about 2000–3000 K with particles of order 10 nm; the Qinghai flicker fits the nearby power line as a modulator, not a source. It is the standard counterpart to the field-held plasmoid of H26 (Force-Free_Harmonic_Plasmoids), which lasts microseconds to milliseconds. A welding arc can make the vapor: the bench test is the electrode fraction vaporized, the particle size and temperature from the spectrum, and silicon-bearing against plain steel electrodes.

Reproducing the results

```
go run ./examples/silicon_ball_lightning -viz examples/silicon_ball_lightning/viz
python examples/silicon_ball_lightning/viz/render.py
python examples/silicon_ball_lightning/viz/make_report.py
go test ./examples/silicon_ball_lightning
```

Sources: J. Abrahamson, J. Dinniss, *Nature* 403, 519 (2000); J. Cen, P. Yuan, S. Xue, *Phys. Rev. Lett.* 112, 035001 (2014); G. S. Paiva et al., *Phys. Rev. Lett.* 98, 048501 (2007); K. D. Stephan, N. Massey, *J. Atmos. Sol.-Terr. Phys.* 70, 1589 (2008); V. Dikhtyar, E. Jerby, *Phys. Rev. Lett.* 96, 045002 (2006); B. E. Deal, A. S. Grove, *J. Appl. Phys.* 36, 3770 (1965); NIST-JANAF tables; CRC Handbook.

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